Assignment: Fundamentals of Linear Algebra for Artificial Intelligence

Instructions

- Answer all the questions.
- Show all intermediate calculations wherever applicable.
- Use a calculator if required.
- Round numerical answers to **two decimal places** where necessary.
- Clearly write the final answer for each question.

Part A – Scalars, Vectors and Matrices

Question 1

Identify whether each of the following is a **Scalar**, **Vector**, **Matrix**, or **Tensor**.

a)

$$5$$

Ans : Scaler

b)

$$\begin{bmatrix} 2 \\ 4 \\ 6 \end{bmatrix}$$

Ans: Vector

c)

$$\begin{bmatrix} 1 & 2 \\ 3 & 4 \end{bmatrix}$$

Ans: Matrix

d)

A colour image of size

$$256 \times 256 \times 3$$

Ans: Tensor

Question 2

Determine the dimensions of the following matrices.

a)

$$\begin{bmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \end{bmatrix}$$

Ans: 2 X 3

b)

$$\begin{bmatrix} 5 \\ 8 \\ 9 \\ 2 \end{bmatrix}$$

Ans: 4 X 1

c)

$$\begin{bmatrix} 1 & 2 \\ 3 & 4 \\ 5 & 6 \end{bmatrix}$$

Ans: 3 X 2

Question 3

Given

$$A = \begin{bmatrix} 2 & 5 \\ 3 & 7 \end{bmatrix}$$

and

$$B = \begin{bmatrix} 4 & 1 \\ 6 & 8 \end{bmatrix}$$

Find

- $A + B$ =[6 6
  9 15]
- $A - B$ =[-2 4
  -3 -1]
- $2A$ =[4 10
  6 14]

Question 4

Find the transpose of the matrix

$$A = \begin{bmatrix} 2 & 4 & 6 \\ 8 & 10 & 12 \end{bmatrix}$$

Ans: [2 8
4 10
6 12]

Question 5

Find the transpose of

$$\begin{bmatrix} 1 & 3 \\ 2 & 5 \\ 4 & 6 \end{bmatrix}$$

Ans: [1 2 4
3 5 6]

Part B – Matrix Multiplication

Question 6

Multiply

$$A = \begin{bmatrix} 1 & 2 \\ 3 & 4 \end{bmatrix}$$

and

$$B = \begin{bmatrix} 5 & 6 \\ 7 & 8 \end{bmatrix}$$

AB= [1 X 5 + 2 X 7 1 X 6 + 2 X 8
3 X 5 + 4 X 7 3 X 6 + 4 X 8]

= [19 22
43 50]

Question 7

Determine whether the following matrix multiplications are possible. If they are possible, write the dimensions of the resulting matrix.

a)

$$(2 \times 3)(3 \times 4)$$

Ans:
Possible. Resulting matrix dimension : 2 X 4

b)

$$(3 \times 2)(4 \times 3)$$

Ans:
Not possible.

c)

$$(4 \times 2)(2 \times 5)$$

Ans:
Possible. Resulting matrix dimension : 4 X 5

d)

$$(5 \times 4)(4 \times 2)$$

Ans:
Possible. Resulting matrix dimension : 5 X 2

Question 8

Given

$$A = \begin{bmatrix} 2 & 1 \\ 4 & 3 \end{bmatrix}$$

and

$$B = \begin{bmatrix} 5 & 6 \\ 7 & 8 \end{bmatrix}$$

Find

$$AB$$

AB = [2 X 5 + 1 X 7   2 X 6 + 1 X 8
4 X 5 + 3 X 7   4 X 6 + 3 X 8]

=[17   20
41   48]

Question 9

Verify whether matrix multiplication is commutative.

Compute

$$AB$$

and

$$BA$$

where

$$A = \begin{bmatrix} 1 & 2 \\ 3 & 4 \end{bmatrix}$$

and

$$B = \begin{bmatrix} 5 & 6 \\ 7 & 8 \end{bmatrix}$$

Based on your calculations, what do you conclude?

Ans:
AB = [1 X 5 + 2 X 7   1 X 6 + 2 X 8
3 X 5 + 4 X 7   3 X 6 + 4 X 8]
= [19   22
43   50]
BA = [5 X 1 + 6 X 3   5 X 2 + 6 X 4
7 X 1 + 8 X 3   7 X 2 + 8 X 4]
= [23   34
31   46]
AB not equal to BA
Therefore matrix multiplication is not commutative.

Part C – Vector Operations

Question 10

Find the magnitude of the vector

$$x = \begin{bmatrix} 3 \\ 4 \end{bmatrix}$$

|x| = square root (3X3 + 4X4)
= 5

Question 11

Find the magnitude of

|y| = square root (2X2 + 5X5 + 6X6)
= 8.0623

$$y = \begin{bmatrix} 2 \\ 5 \\ 6 \end{bmatrix}$$

Question 12

Compute the dot product of

$$x = \begin{bmatrix} 2 \\ 4 \\ 1 \end{bmatrix}$$

x.y = 2X5 + 4X2 + 1X3
= 21

and

$$y = \begin{bmatrix} 5 \\ 2 \\ 3 \end{bmatrix}$$

Question 13

Determine whether the following vectors are orthogonal.

$$u = \begin{bmatrix} 2 \\ 1 \end{bmatrix}$$
$$v = \begin{bmatrix} 1 \\ -2 \end{bmatrix}$$

Ans: If the vectors u, v are orthogonal then u.v = 0
here, u.v = 2X1 + (1X-2)
= 0
hence the victors u and v are orthogonal.

Question 14

Normalize the vector

$$x = \begin{bmatrix} 6 \\ 8 \end{bmatrix}$$

Ans: ||x|| = square root (6X6 + 8X8)
= 10
normalize vector = [0.6
0.8]

Question 15

Find the Euclidean distance between

$$A = (2, 3)$$

and

$$B = (8, 11)$$

Ans: = square root ((8-2)X(8-2) + (11-3) X(11-3))
= square root (36+64)
= 10

Part D – Vector Spaces and Span

Question 16

Determine whether the following vectors are linearly independent.

$$v_1 = \begin{bmatrix} 1 \\ 2 \end{bmatrix}$$
$$v_2 = \begin{bmatrix} 2 \\ 4 \end{bmatrix}$$

Ans: V1 and V2 are not linearly independent, because
V2 = 2V1

Explain your answer.

Question 17

Determine whether

$$v = \begin{bmatrix} 5 \\ 7 \end{bmatrix}$$

belongs to the span of

$$v_1 = \begin{bmatrix} 1 \\ 1 \end{bmatrix}$$

and

$$v_2 = \begin{bmatrix} 2 \\ 1 \end{bmatrix}$$

Ans: span(v1, v2) = {av1+bv2}
a and b are any real numbers.
v = av1+bv2
5 = a+2b
7 = a+b
a = 9 and b = -2
hence,

v= 9v1 + (-2)v2
(a possible linear combination od v1 and v2)

v belongs to the span of v1 and v2.

Question 18

Find a basis for

$$\mathbb{R}^2$$

Ans: [1
0]
and
[0
1]

Question 19

Find the rank of the matrix

$$\begin{bmatrix} 1 & 2 \\ 2 & 4 \end{bmatrix}$$

Ans: Row 2 = 2Row1
(Row2 is dependent of Row1)
so,

rank = 1

Question 20

Find the rank of

$$\begin{bmatrix} 1 & 2 \\ 3 & 4 \end{bmatrix}$$

Ans:
Rows are independent.
Rank = 2

Part E – Orthogonality

Question 21

State whether the following pairs of vectors are orthogonal.

a)

$$\begin{bmatrix} 1 \\ 0 \end{bmatrix}, \begin{bmatrix} 0 \\ 1 \end{bmatrix}$$

Ans : Dot product = 0
vectors are orthogonal.

b)

$$\begin{bmatrix} 3 \\ 4 \end{bmatrix}, \begin{bmatrix} 4 \\ -3 \end{bmatrix}$$

Ans : Dot product = 0
vectors are orthogonal.

c)

$$\begin{bmatrix} 2 \\ 5 \end{bmatrix}, \begin{bmatrix} 5 \\ 2 \end{bmatrix}$$

Ans : Dot product = 10
vectors are not orthogonal.

Question 22

Find the cosine similarity between

$$x = \begin{bmatrix} 1 \\ 2 \end{bmatrix}$$

and

$$y = \begin{bmatrix} 3 \\ 4 \end{bmatrix}$$

Interpret the result.

Ans:
cosine similarity = (x y)/ ||x|| ||y||
x.y = 1X3 + 2X4 = 11
||x|| = square root (1X1 + 2x2) = 2.236
||y|| = square root (3X3 + 4X4) = 5

cosine similarity = 11/ (2.236 X 5)
= 11/11.18
= 0.983
Interpretation:
Cosine similarity is close to +1.
Means, the vectors x and y are in similar or identical directions.

Question 23

Explain, in your own words, the difference between:

- Orthogonal vectors
- Orthonormal vectors
- A basis
- A vector space

Orthogonal vectors:
when two vectors are said to be orthogonal if they are perpendicular to each other with any magnitude. That is,
1. the dot products equals zero
2. they form 90-degree angle i.e. directions are completely different or opposite.
3. represent independent information.

orthonormal vectors:
when two vectors are orthonormal when,
1. they are orthogonal vectors and
2. their magnitude is one (unit length) to remove the bias i.e. normalizing vectors.

A basis:
A basis is the set of linear independent vectors that span the entire vector space.
the basis represents the minimum number of vectors to describe each vector in that space i.e. it holds minimum independent information.

A vector space:
A vector space is collection of all vectors in every possible direction such that adding vectors and multiplying vectors by numbers stays inside that space. i.e. it holds all the information some of which are redundant.